

05 The Rate Constant

Why are we learning this?

In the previous lecture, we learned that reaction rates depend on concentration through the rate law. However, reaction rates also depend strongly on temperature.

Today we will study the **rate constant**, k , and learn why reactions proceed much more quickly at higher temperatures. By the end of today's lesson, you should understand:

- What the rate constant represents.
- How the Arrhenius equation relates temperature and reaction rate.
- The physical meaning of the frequency factor.
- The physical meaning of activation energy.
- Why reaction rates often increase dramatically with temperature.
- How temperature influences reactor design.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



► Rate Laws ◀
(Kinetics)



Equilibrium

(Reversible Reactions)



Equilibrium Conversion

(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



One Reaction: Reactor Design

(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods

(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption

(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption

(Reaction Mechanisms)

Big Idea

Reaction rates depend on:

Concentration

+

Temperature

Today we learn how temperature enters the rate expression.

We saw in the last lecture that the rate law is equal to a rate constant k multiplied by some function of concentration. For example: $-r_A = k \cdot f(C_A, C_B, C_C, \dots)$. In this lecture, we'll concern ourselves with the rate constant k .

Why Study the Rate Constant?

The concentration term in the rate law tells us how reaction rate changes when concentrations change.

The rate constant tells us how reaction rate changes when temperature changes.

For many reactions, temperature has a much larger effect on reaction rate than concentration.

This is why temperature control is one of the most important aspects of reactor design.

The rate constant follows an Arrhenius equation: $k_A(T) = A(T)\exp(-\frac{E}{RT})$, where $A(T)$ is the pre-exponential term or "frequency factor" and E is the activation energy.

$A(T)$ quantifies

$\exp(-\frac{E}{RT})$ is

Specifically, $\exp(-\frac{E}{RT})$ is a Boltzmann probability, which is derived from statistical mechanics.

What Does the Arrhenius Equation Tell Us?

The Arrhenius equation relates the rate constant to temperature.

It tells us:

- Faster reactions have larger values of k .
- Increasing temperature increases k .
- Reactions with larger activation energies are more sensitive to temperature changes.

This equation is one of the most important equations in chemical reaction engineering.

Example: In the gas phase, reactions happen when gas molecules collide. However, not every collision results in a reaction; only collisions with sufficient energy to surpass the activation energy result in a reaction. $A(T)$ quantifies the total the frequency of collisions, i.e., the number of collisions per unit time, and even accounts for those collisions that do not have sufficient energy to surpass the activation energy. The exponential term $\exp(-\frac{E}{RT})$ scales $A(T)$ by that fraction of collisions that have sufficient energy to surpass the activation energy.

Gas molecules zoom around really fast, sometimes they collide. The frequency of collisions can be predicted using theorems from physical chemistry.

E is the activation energy. It is equal to the difference in enthalpy between the transition state and the reactants.

The transition state is a compound with chemical structure "in between" that of the reactants and products. For example, for the isomerization reaction $\text{HCN} \rightarrow \text{CNH}$, the transition state looks like:

Notice how the transition state is "contorted" compared to the reactants and products. This contortion costs energy ... this energy cost gives rise to the activation energy. We can illustrate these energies on a "potential energy surface":

Two Factors Control the Reaction Rate Constant

The Arrhenius equation contains two important pieces:

- 1. Frequency Factor A :** Measures how often molecules encounter one another in ways capable of leading to reaction.
- 2. Exponential Term $e^{-E/RT}$:** Measures the fraction of molecular encounters with sufficient energy to react.

Both factors must be favorable for a reaction to proceed rapidly.

Why Doesn't Every Collision Produce a Reaction?

Molecules collide constantly. Most collisions do **not** produce a reaction. For a reaction to occur, molecules must possess enough energy to reach the transition state. The activation energy represents this energetic barrier. Only some molecular collisions successfully cross this barrier.

What Is a Transition State?

A transition state is a short-lived arrangement of atoms that exists between reactants and products. It is not a stable chemical species. Instead, it represents the highest-energy point along the reaction pathway. Because the transition state is unstable, reaching it requires energy, giving rise to the activation energy.

Why Does Temperature Have Such a Large Effect?

Temperature appears inside an exponential function:

$$e^{-E/RT}$$

Small changes in temperature can therefore produce very large changes in reaction rate. This is why:

- Food spoils more quickly at room temperature than in a refrigerator.
- Crickets chirp more frequently when it's hotter outside.
- Temperature control is critical for reactor safety.

Summary and Key Takeaways

Today we examined the rate constant and its temperature dependence.

Key Ideas

- ✓ Reaction rates depend on both concentration and temperature.
- ✓ The Arrhenius equation, $k = Ae^{-E/RT}$, describes how temperature affects the rate constant.
- ✓ The frequency factor describes how often potentially reactive encounters occur.
- ✓ The activation energy represents the energetic barrier that must be overcome.
- ✓ Only a fraction of molecular collisions have sufficient energy to react.
- ✓ Temperature has a large effect because it appears inside an exponential function.

Most Important Idea

Concentration affects rate through the rate law. Temperature affects rate through the rate constant. This connection between temperature and reaction rate will become central when we study energy balances and non-isothermal reactor design.